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Education

The University of Alabama in Huntsville (UAH) - M.S.E Electrical Engineering	2024 – 2025
Iowa State University (ISU) - B.S. Mechanical Engineering	2020 – 2023

Experience

Graduate Research Assistant <i>Nonlinear and Complex Systems Lab (NACS-Lab)</i>	<i>Huntsville, AL</i> 2024 - present
- Machine Learning and Data-Driven Modeling (Python, MATLAB, PyTorch) - Wrote custom reservoir computing and sparse identification of nonlinear dynamics implementations to analyze real-world chaotic oscillators. These systems present a significant modeling challenge because they are mathematically proven to be random. - Fiber Amplifier Laser Enhancement (MATLAB, Python, VISA) - Applied machine learning to semiconductor diodes simulations to analyze an intractable system. - Radar Waveform Design (C++, CMake, GNU Radio, Docker, FFTW, Cuda) - Wrote custom C++ to achieve 400x performance increase for calculation/display of radar ambiguity surface. High speed display allowed for improved analysis of novel noise radar methods.	
Machine Learning Engineering Intern <i>General Atomics Aeronautical Systems inc.(GA-ASI)</i>	<i>Poway, CA</i> Summer 2023
- Multi-Agent Reinforcement Learning Training Enhancement (Python, C#, Unity3D, MLAgents) - Developed deterministic controller to improve multi-agent training environment to enable safe effective autonomy operationalization. - Utilized controller to begin benchmarking of reinforcement learning agent, helping analyze which agents would deploy most effectively. - Development Research (Unity3D, MLAgents) - Performed stack tracing to discover and highlight problems in communication backend. Helped optimize software for speed and throughput to produce results faster. - Began preliminary literature search to expand knowledge base in state-of-the-art robustness methods allowing for more secure model development.	
Undergraduate Research Assistant <i>Virtual Reality Applications Center (VRAC)</i>	<i>Ames, IA</i> 2020 - 2023
- Mixed Reality Demonstration Applications (Unity3D, C#, Arduino, Oculus) - Designed and built passive haptic feedback system for mixed reality application that demonstrated enhanced user experience and immersion through tactile sensation. - Stood up Unity project for immersive carrier application for cave automatic virtual environment that highlighted the unique system capabilities to potential clients.	

Publications

(Poster) “Real-World Data-Driven Modeling for Chaotic Circuits and Nonlinear Elements” Mattan Tseng, Aubrey Beal, Scott Meadows, Daniel Matyas	<i>(Accepted) May. 2025</i>
(Conference) “Enhanced suppression of stimulated Brillouin scattering (SBS) in high energy fiber lasers through novel chaotic phase modulation” Daniel Matyas, Mattan Tseng, Zachary C Helton, Christopher N Vincent, Aubrey Beal	<i>(Accepted) Apr. 2025</i>
(Journal) “XRAE: Extended Reality Acuity Examination” Mattan Tseng, Jack Miller, Dante Goldner, Eliot Winer, https://doi.org/10.1016/j.softx.2023.101429	<i>Jul. 2023</i>

Skills

Languages: Python, MATLAB, C#, C++
Technologies: git, Unity Game Engine, Docker, Numpy, Pandas, PyTorch, SolidWorks